

# Transmission Media

Module 22

Duration: 26:57

Video: <https://www.youtube.com/watch?v=Xw7oyDuqOwU>

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0:00 [Music]

0:16 my dear students hello I welcome you all in our series of lectures on cyber security today we are going to study

0:25 about transmission media we will also learn how to measure the performance of various transmission

0:33 Medias so let us begin with it transmission media is a pathway that carries the information from sender to

0:41 receiver we use different types of cables to transmit data data is transmitted normally through electrical

0:48 or electromagnetic signals an electrical signal is in the form of current an electromagnetic signal is series of electromagnetic energy pulses at various

0:57 frequencies these signals can be transmitted through copper wires Optical fibers atmosphere water and vacuum

1:05 different medias have different properties like bandwidth delay cost and ease of installation and maintenance transmission media is also called

1:14 Communication channel the figure shows the position of transmission media in the relation to a physical layer now what are the various factors

1:22 for design of data transmission systems a key concern generally is data rate and distance the the greater the data rate

1:31 and distance the better a number of design factors relating to the transmission medium and to the signal determine the data rate and the distance

1:39 such as number one band width all other factors remaining constant the greater the bandwidth of a signal the higher the

1:47 data rate that can be achieved number two transmission impairment impairments such as

1:56 attenuation limit the distance for guided media Twisted paer generally suffer more impairment than coxl cable

2:03 which in turn suffers more than optical fiber number three interference interference from competing

2:13 signals in overlapping frequency bands can distort or wipe out a signal interference is of particular

2:21 concern for unguided media but it is also a problem with guided media proper shielding of a guided

2:29 medium can minimize this problem number four number of receivers a guided medium can be used to

2:37 construct a point-to-point link or a sheer link with multiple attachments in the later case each attachment introduces some attenuation

2:46 and distortion on the line limited distance and the data rate next is types of transmission media

2:55 the transmission media is categorized as number one guided media for guided transmission media the transmission

3:04 capacity in terms of either data rate or bandwidth depends critically on the distance and on whether the medium is

3:12 point to point or multi-point the three guided media commonly used for data transmission are

3:19 twisted pair coaxial cable and optical fiber let us now learn about twisted pair a twisted pair consists of two

3:28 conductors normally copper each with its own plastic insulation one of the wires is used to

3:36 carry signals to the receiver and the other is used only as a ground reference the receiver uses the

3:43 difference between the two and addition to the signal sent by the sender on one of the wires interference and cross talk

3:52 may affect both wires and create unwanted signals if the two wires are parallel the effect of these unwanted

3:59 signal is not the same in both wires because they are at different locations relative to the noise or cross talk

4:07 sources this results in a difference at the receiver by twisting the pairs a balance is maintained for example

4:15 suppose in one twist one wire is closer to the noise source and the other is farther in the next twist the reverse is

4:23 true twisting makes it probable that both wires are equally affected by external influences such as noise or

4:31 cross talk this means that the receiver which calculates the difference between the two receives no unwanted signals The

4:39 Unwanted signals are mostly canceled out the number of twist per unit of length called has some effect on the quality of

4:46 the cable twisted pair is of two types unshielded twisted pair cable and shielded twisted pair

4:56 cable the most common twisted pair cable used in Communications is referred to as unshielded Twisted pair

5:03 cable shielded Twisted pair cable has a metal foil or braided mesh covering that encases each pair of insulated

5:12 conductors although metal casing improves the quality of the cable by preventing the penetration of noise or

5:18 cross talk it is bulkier and more expensive although UTP was once considered to be slower at transmitting

5:27 data than other types of cable this is no longer true in fact UTP is considered the

5:34 fastest copper based medium today the main features of unshielded Twisted cable are number one speed and

5:43 throughput is from 10 to 1000 Mbps number two average cost per node is

5:50 very less and number three media and connector size used with it is very

5:57 small next is connectors the most common UTP connector is RJ45 where RJ stands for registered Jack then

6:06 comes shielded twisted pair cable shielded twisted pair cable combines the techniques of shielding cancellation and wire

6:14 twisting each pair of wires is wrapped in a metallic foil the four pairs of wires then are wrapped in an overall

6:22 metallic braid or foil usually 150 ohm cable as specified for use in ether

6:29 Network installations shielded Twisted pair reduces electrical noise both within the cable and from outside the

6:39 cable Shield twisted pair usually is installed with STP data connector which is created especially for the STP cable

6:48 however STP cabling also can use the same RG connectors that UTP

6:55 uses although STP prevents interference better than UTP it is more expensive and difficult to install in addition the

7:04 metallic shielding must be grounded at both ends if it is improperly grounded The Shield acts like an antenna and picks up unwanted signals because of its cost and

7:11 difficulty with termination STP is rarely used in Ethernet networks STP is primarily used in Europe the main features

7:18 of STP cable are number one speed and throughput is from 10 to 100000 Mbps

7:27 number two average cost per node is moderately High number three media and

7:34 connector

7:41 size used with it varies from medium to large next is applications of Twisted

7:49 pairs there are many applications such as UTP cables are found in many ethernet

7:56 networks UTP cables are used in telephone systems it is the most commonly used medium for Workhorse for communications

8:05 within buildings in the telephone system individual residential telephone sets are

connected to the local telephone

8:12 exchange or end office by twisted pair wire these are referred to as subscriber Loops there are many advantages

8:21 associated with twisted pair cables such as their installation is easy they're flexible and cheap and they have a high

8:29 speed capacity the disadvantage is that their bandwidth is low when compared with coaxial cable and they provide less

8:38 protection from interference next is coaxial cables coaxial is called by this name

8:46 because it contains two conductors that are parallel to each other copper is used in this a center

8:54 conductor which can be a solid wire or a stranded one it is surrounded by PVC installation

9:01 9 minutes, 1 second a sheath which is encased in an outer conductor of metal foil outer metallic wrapping is used as

9:09 a shield against noise and the second conductor which completes the circuit the outer conductor is also

9:18 encased in an insulating sheet the outermost part is the plastic cover which protects the whole

9:26 cable the most common coaxial standards are number one 50 ohm RG7 or RG11 which is

9:34 used with thick ethernet number two 50 ohm RG58 used

9:41 with thin ethernet number three 75 ohm RG59 which is used with cable television

9:48 and number four 93 ohm RG62 used with Airtel the figure shows the coaxial cable

9:57 next is coaxial cable connectors to connect coaxial cable to devices we need coaxial

10:04 connectors the most common type of connector used today is the BNC connector the BNC T connector and the BNC

10:13 Terminator the BNC connector is used to connect the end of the cable to a device such as a TV

10:21 set the BNC T connector is used in Ethernet networks to branch out to a connection to a computer or other device

10:30 device the BNC Terminator is used at the end of the cable to prevent the reflection of the signal now what about the performance of

10:39 the coaxial cables the attenuation is much higher in coaxial cables than in the twisted pair

10:45 cables although coaxial cable has a much higher bandwidth the signal weakens rapidly and requires the frequent use of

10:54 repeaters next is various applications of coaxial cables number one coaxial cable was widely used in analog telephone networks

11:02 where a single coaxial Network could carry 10,000 voice signals number two it was

used in digital telephone networks where

- 11:11 a single coaxial cable could carry Digital Data up to 600 Mbps however coaxial Cable
- 11:19 in telephone networks has largely been replaced today with fiber optic cables
- number three cable TV networks
- 11:27 also use coaxial cables fourth because of its high bandwidth and consequently High
- data rate coaxial cable
- 11:35 was chosen for digital Transmission in early ethernet lens next is various types of
- coaxial cables there are two types
- 11:43 of coaxial cables baseband and Broadband baseband coaxial cable is a 50
- 11:49 ohm coaxial cable which is used for digital transmission it is mostly used for
- 11:57 lens Baseband transmits a single signal at a time with very high speed the major
- drawback is that it needs amplification
- 12:05 after every 1000 ft Broadband coaxial cable uses analog transmission on standard
- cable television cabling it
- 12:13 transmits several simultaneous signals using different frequencies it covers large
- area when compared with baseband coaxial
- 12:22 cables it has many advantages such as number one bandwidth is high number two
- 12:29 used in longdistance telephone lines number three transmits digital signals at a
- very high rate of 10
- 12:37 mpps number four has much higher noise immunity number five performs data
- 12:44 transmission without Distortion and number six they can span to longer distance at
- higher speeds as
- 12:52 they have better shielding when compared to Twisted pair cables it has some
- disadvantages also such as
- 13:00 number one single cable failure can fail the entire network number two difficult to
- install
- 13:07 and expensive when compared with twisted pair and number three if the shield is
- imperfect it can lead to grounded
- 13:17 Loop next is fiber optic cable a fiber optic cable is made of glass or plastic and
- transmit signals in
- 13:26 the form of FL figure shows the optical fiber light travels in a straight line
- 13:33 as long as it is moving through a single uniform substance if a ray of light traveling
- through one substance suddenly
- 13:40 enters another substance of a different density the ray changes direction as the
- figure shows if the
- 13:48 angle of incidence  $I$  which is the angle the Ray makes with the line perpendicular
- to the interface between
- 13:56 the two substances is less than the critical angle the ray reflects and moves closer
- to the

- 14:03 surface if the angle of incidence is equal to the critical angle the light bends along the
- 14:10 interface if the angle is greater than the critical angle the ray reflects back and travels again in the denser
- 14:18 substance note that the critical angle is a property of the substance and its value differs from one substance to
- 14:26 another Optical fibers use refraction to guide light through a channel figure shows the bending of a
- 14:34 light Ray fiber optic cable has a bandwidth of more than 2 GB per
- 14:42 second next is propagation modes current technology supports two modes called multi mode and single mode
- 14:51 for propagating light along Optical channels each requiring fiber with different physical characteristics
- 14:59 multimode can be implemented in two forms step index or graded index multiple beams from a light source
- 15:07 move through the core in different paths how these beams move within the cable depends on the structure of the
- 15:15 core in multimode step index fiber the density of the core remains constant from the center to the edges a beam of
- 15:23 light moves through this constant density in a straight line until it reaches the interface of the core and the
- 15:31 cladding at the interface there is an abrupt change due to a lower density this alters the angle of the beam's
- 15:40 motion the term step index refers to the suddenness of this change which contributes to the distortion of the
- 15:47 signal as it passes through the fiber a second type of fiber called multimode graded index fiber decreases
- 15:56 this distortion of the signal through the core the word index here refers to the index of
- 16:04 refraction the index of refraction is related to density density is highest at the center of the core and decreases gradually to
- 16:13 its lowest at the edge single mode uses step index fiber and a highly focused source of light that
- 16:21 limits beams to a small range of angles all close to the horizontal figure shows various fibers
- 16:29 optic transmission modes next is fiber optic cable connectors there are three types of
- 16:37 connectors for fiber optic cables the subscriber channel connector is used for cable TV it uses a push pull

16:46 Locking System the straight type connector is used for connecting cable to network devices it uses a Bonet

16:53 Locking System and is more liable than SC Mt RJ is another connector that is

17:00 used and is the same size as RJ45 next comes applications of fiber optic cables first application is fiber

17:10 optic cable is often found in backbone networks because it's wide bandwidth is cost

17:17 effective number two some cable TV companies use a combination of optical fiber and coxl cable thus creating a

17:25 hybrid Network optical fiber provides the backbone structure while coxl cable provides the connection to the user

17:32 premises third application is local area network such as 100 base FX Network and

17:38 thousand base X also uses Fiber Optic Cables there are many advantages

17:45 associated with fiber optics such as number one they provide high quality transmission of signals at very high

17:52 speed number two these are not affected by electromagnetic interference so noise and Distortion is very less l number

17:59 three they are used for both analog and digital signals they have some disadvantages too such as number one

18:07 they're expensive number two difficult to install number three maintenance is expensive and difficult and number four

18:16 they do not allow complete rooting of light signals next is unguided media that is Wireless unguided metor transport

18:25 electromagnetic waves without using a physical conductor this type of communication is often referred to as a wireless communication

18:34 signals are normally broadcast through free space and thus are available to anyone who has a device capable of receiving

18:43 them unguided signals can travel from the source to the destination in several ways such as ground propagation Sky

18:51 propagation and line of sight propagation figure shows various propagation methods

18:59 the section of the electromagnetic spectrum defined as radio waves and microwaves is divided into eight ranges called bands each regulated by

19:08 government authorities these bands are rated from very low frequency to extremely high

19:16 frequency for unguided media transmission and reception are achieved by means of an antenna an antenna can be

19:24 defined as an electrical conductor or system of conductors used used either for radiating electromagnetic energy or

19:31 for collecting electromagnetic energy for transmission of a signal electrical energy from the transmitter

19:38 is converted into electromagnetic energy by the antenna and radiated into the surrounding environment like atmosphere

19:45 space or water for reception of a signal electromagnetic energy impinging on the antenna is converted into electrical

19:54 energy and fed into the receiver in Satellite telecommunication a down link is the link from a satellite

20:03 down to one or more ground stations or receivers and an Uplink is a link from a ground station up to a

20:11 satellite some companies sell Uplink and downlink services to television stations corporations and to other

20:18 telecommunication carriers a company can specialize in providing uplinks down links or both we can divide Wireless transmission

20:27 into three Broad groups radio waves microwaves and infrared waves two separate frequencies are used so that

20:36 one cannot interfere with the other and full duplex communication is possible and other reason is that the

20:43 power required to transmit a signal is proportional to the frequency of the signal the more the Power requirement more would be the weight of the

20:52 system next is radio waves electromagnetic waves ranging in the frequency between 3 khz and 1 GHz

21:02 are normally called radio waves radio waves are om directional when an antenna transmit radio waves

21:09 they are propagated in all directions this means that the sending and receiving antennas do not have to be

21:17 aligned a sending antenna sends waves that can be received by any receiving antenna the OM property has a

21:24 disadvantage too the radio waves transmitted by one antenna are susceptible to interference

21:31 by another antenna that may send signals using the same frequency or band radio waves particularly those waves that

21:39 propagate in the sky mode can travel long distances this makes radio waves a good candidate for long-distance

21:46 broadcasting such as AM radio radio waves particularly those of low and medium frequencies can penetrate walls

21:54 this characteristic can be both an advantage and a disadvantage it is an advantage because for example an AM

22:01 22 minutes, 1 second radio can receive signals inside a building it is a disadvantage because we cannot isolate a communication to just

22:09 inside or outside a building the radio wave band is relatively narrower just under 1 GHz

22:16 compared to the microwave band now various applications of radio waves the only Direction characteristics of radio

22:24 waves make them useful for multicasting in which there is one sender but many receivers AM and FM radio

22:31 television Maritime radio carless phones and paging are examples of multicasting next is

22:41 microwaves electromagnetic waves having frequencies between 1 and 3 GHz are called microwaves microwaves are

22:48 unidirectional when an antenna transmits microwaves they can be narrowly focused this means that the sending and

22:56 receiving antennas need to be aligned the unidirectional property has an obvious Advantage a pair of antennas can be

23:04 aligned without interfering with another pair of aligned antennas there are some characteristics of

23:10 microwave propagation such as number one microwave propagation is line of sight since the towers with the mounted

23:18 antennas need to be in direct sight of each other towers that are far apart need to be very

23:25 tall the curvature of the earth as well as other blocking obstacles do not allow two short towers to communicate by using

23:33 microwaves repeaters are often needed for long distance Communications number two very high frequency microwaves

23:41 cannot penetrate walls this characteristic can be disadvantage if receivers are inside buildings number three the microwave

23:50 band is relatively wide almost 299 GHz therefore wider subbands can be assigned and a high data rate is possible

23:59 and number four use of certain portions of the band requires permissions from authorities now applications of

24:07 microwaves microwaves due to their unique properties are very useful when unicast communication is needed between

24:15 the sender and the receiver they're used in cellular phones satellite networks and Wireless LAN last but not the least is

24:23 infrared waves infrared waves with frequencies from 300 GHz to 400 THz can be used

24:32 for short range communication infrared waves having high frequencies cannot penetrate walls this advantageous characteristics

24:41 prevents interference between one system and another a short range communication system in one room cannot be affected by

24:49 another system in the next room when we use our infrared remote control we do not interfere with the use of the remote

24:56 by our neighbors however this same characteristic makes infr signals useless for longrange

25:05 Communications in addition we cannot use infrared waves outside a building because the sun's Rays contain infrared

25:12 waves that can interfere with the communication now let us discuss various applications of infrared waves infrared

25:19 band almost 400 trz has an excellent potential for data transmission such a wide bandwidth can

25:27 be used to transmit Digital Data with a very high data rate the infrared data Association and Association for sponsoring the use of

25:35 infrared waves has established standards for using these signals for communications between devices such as

25:43 keyboards Mouse PCS and printers for example some manufacturers provide a special Port called the irda port that

25:52 allows a wireless keyboard to communicate with a PC the standard originally defined a dat rate of 75

25:59 kilobits per second for a distance up to 8 m the recent standard defines a data rate of 4

26:07 MVPs so with this our today's lecture is over today we discussed about various transmission media and their features we

26:16 also discussed in detail the advantages and disadvantages associated with them I hope you have understood the concepts in

26:24 the next lecture we will discuss about Wireless networking till then goodbye thank you have a nice day

26:51 [Music]